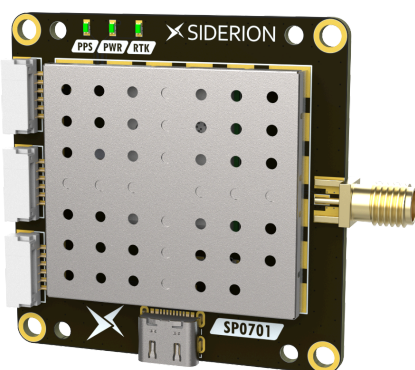


*Delivers centimeter level accuracy within seconds based on U-blox triple-band L1/L2/L5 **ZED-F20P** GNSS RTK module.*

*Datasheet - In Production*



## Features

- 5V - 100mA power supply
- LEDs status: Power/RTK
- Timepulse & External Interrupt
- USB-C/I<sup>2</sup>C/SPI/UART interfaces
- Gold plated SMA connector
- 3.3V Active antenna input

## Applications

- Drones
- RTK Base
- Ground vehicles
- Precise navigation

## Description

The Siderion **SP0701** is an affordable, compact, and high-precision triple-band L1/L2/L5 GNSS RTK (Real-Time Kinematic) device.

Powered by the latest generation of u-blox **ZED-F20P** modules, the SP0701 supports concurrent reception of GPS, Galileo, BeiDou, QZSS, and NavIC signals. Consequently, the device delivers reliable performance and fast convergence, achieving centimeter-level accuracy in seconds.

Its JST-GH connectors make it ideal for integration with Pixhawk or other autopilot systems.

The SP0701 operates across a temperature range from -40°C to +70°C.

Table 1. Device summary

| Order ref code      | Temperature [°C] | Product size [mm] | Weight [g] |
|---------------------|------------------|-------------------|------------|
| SID-GNSS-RTK-SP0701 | -40 to +70       | 50.0 x 50.0 x 7.8 | 17         |

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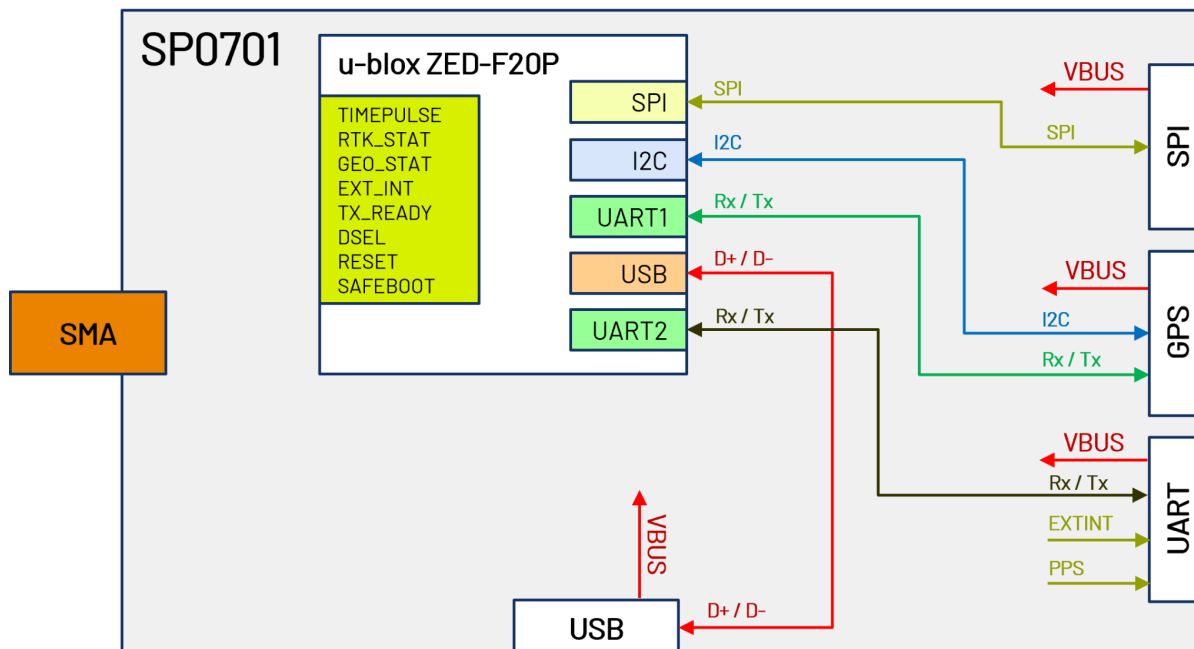
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# 1. Block diagram and pin description

## 1.1 Block diagram

Figure 1. SP0701 block diagram & connectivity



## 1.2 LED description

Table 2. SP0701 LED sequence status

| LED name   | Color | Light sequence | Comment                              |
|------------|-------|----------------|--------------------------------------|
| Timepulse  | Green |                | Blinking LED at the defined PPS rate |
| Power      | Blue  |                | Solid blue LED when powered ON       |
| RTK Status | Green |                | 3D fix mode / No RTK fix             |
|            |       |                | RTK fix but no FIXED RTK fix         |
|            |       |                | FIXED RTK fix                        |

## 1.3 Pin description

Figure 2. SP0701 pinout (top view)

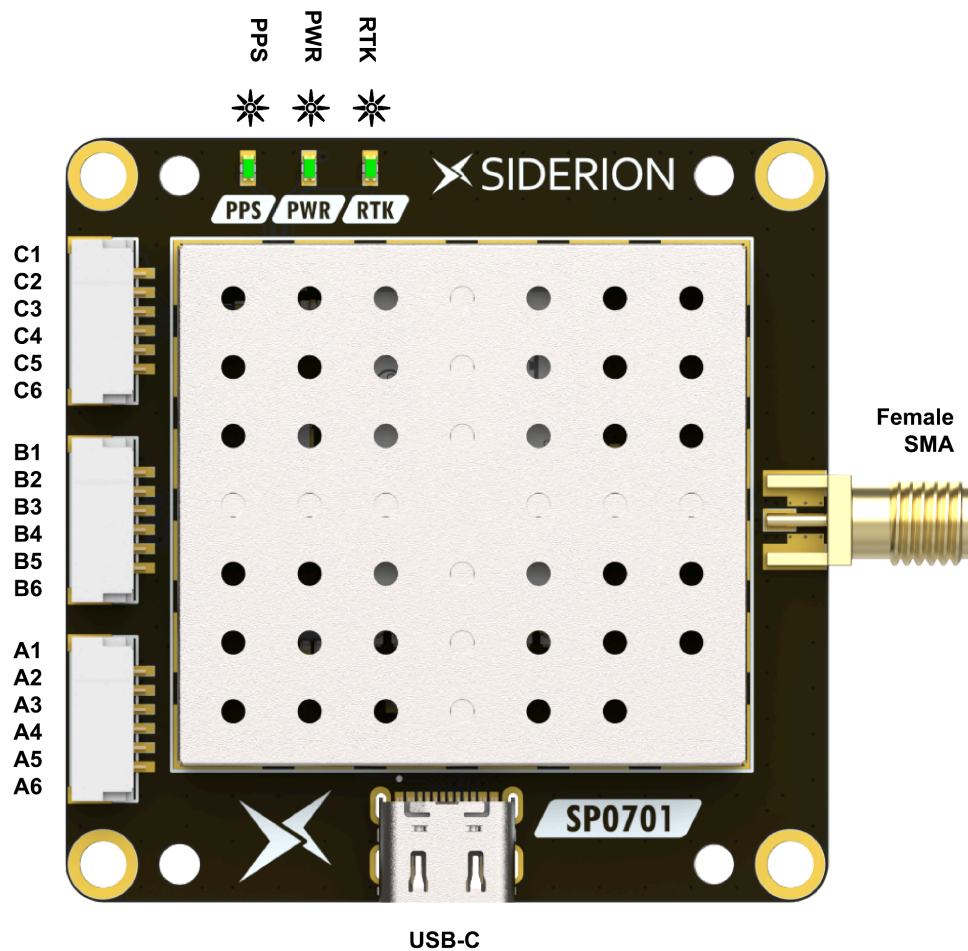


Table 3. Pinout configuration

|     | Pin | Name     | Type | Function                                |
|-----|-----|----------|------|-----------------------------------------|
| SPI | A1  | 5V SPI   | P    | 5V input to select SPI instead of UART1 |
|     | A2  | SPI CLK  | I    | SPI clock input                         |
|     | A3  | SPI MISO | I    | SPI Master Input Slave Output           |
|     | A4  | SPI MOSI | O    | SPI Master Output Slave Input           |
|     | A5  | SPI CS   | I    | SPI chip select input                   |
|     | A6  | GND      | P    | Ground reference                        |

P : Power / I : Input / O : Output

|        | Pin | Name                 | Type | Function                           |
|--------|-----|----------------------|------|------------------------------------|
| GPS    | B1  | 5V IN                | P    | 5V input                           |
|        | B2  | UART1 RXD / SPI MOSI | I    | UART1 receive <b>OR</b> SPI MOSI   |
|        | B3  | UART1 TXD / SPI MISO | O    | UART1 transmit <b>OR</b> SPI MISO  |
|        | B4  | I2C SCL / SPI SCLK   | I    | I2C clock <b>OR</b> SPI clock      |
|        | B5  | I2C SDA / SPI CS     | I/O  | I2C data <b>OR</b> SPI chip select |
|        | B6  | GND                  | P    | Ground reference                   |
| UART 2 | C1  | 5V IN                | P    | 5V input                           |
|        | C2  | UART2 TXD            | O    | UART2 transmit NMEA/RTCMdata       |
|        | C3  | UART2 RXD            | I    | UART2 receive NMEA/RTCMdata        |
|        | C4  | EXTINT               | I    | External interrupt                 |
|        | C5  | TIMEPULSE            | O    | External interrupt on Timepulse    |
|        | C6  | GND                  | P    | Ground reference                   |

**Note:** The SPI port is available only when the **D\_SEL** pin (located underneath) is pulled LOW.

## 2. Absolute maximum ratings

Stresses above those listed as “absolute maximum ratings” may cause permanent damage to the device. This is a stress rating only and functional operation of the device under these conditions is not implied. Exposure to maximum rating conditions for extended periods may affect device reliability.

Table 4. SP0701 absolute maximum ratings

| Symbol | Parameter               | Maximum value   | Units |
|--------|-------------------------|-----------------|-------|
| Vusb   | USB supply voltage      | -0.3 to +6      | V     |
| Vdd    | Internal supply voltage | -0.5 to +3.6    | V     |
| Vdd_IO | I/O pins supply voltage | -0.5 to Vdd+0.1 | V     |
| Icc_RF | RF output current       | 300             | mA    |
| Prfin  | Input power at RF_IN    | 10              | dBm   |
| W      | Product weight          | 17              | g     |
| TOP    | Operating temperature   | -40 to +70      | °C    |
| TSTG   | Storage temperature     | -40 to +80      | °C    |



This device is sensitive to electrostatic discharge (ESD), improper handling can cause permanent damage to the part.

## 3. Specifications

### 3.1 Ublox ZED-F20P specifications

Table 5. ZED-F20P mechanical and electrical specifications

| Symbol | Parameter                             | Test conditions    | Min.    | Typ.    | Max.     | Units |
|--------|---------------------------------------|--------------------|---------|---------|----------|-------|
| Vusb   | USB supply voltage                    |                    | 4.5     | 5.0     | 5.5      | V     |
| Vdd    | Internal supply voltage               |                    |         | 3.3     |          | V     |
| Vdd_IO | Supply voltage for I/O                |                    |         | 3.3     |          | V     |
| Idd    | Current consumption                   | w/o active antenna |         | 50      |          | mA    |
|        |                                       | w/ active antenna  |         | 130     |          | mA    |
| Vdd_RF | Output RF voltage                     |                    |         | Vdd-0.1 |          | V     |
| Idd_RF | Output RF current                     |                    |         | 50      |          | mA    |
| Vil    | IO pin low lvl input volt             |                    | 0       |         | 0.8      | V     |
| Vih    | IO pin high level input voltage       |                    | 2       |         | Vdd +0.3 | V     |
| Vol    | IO pin low level output voltage       | Iol = 1mA          |         |         | 0.4      | V     |
| Voh    | IO pin high level output voltage      | Ioh = 1mA          | Vdd-0.4 |         |          | V     |
| Top    | Operating temperature                 |                    | -40     |         | +70      | °C    |
| I2Cadd | Default 7bit I <sup>2</sup> C address | 0b0100010 (0x42)   |         |         |          |       |

Table 6. ZED-F20P general performance

| Parameter              | Specifications                   | Value                       |
|------------------------|----------------------------------|-----------------------------|
| Receiver type          | Multi-band GNSS high precision   |                             |
| Accuracy of Timepulse  | RMS                              | 30 ns                       |
| Frequency of Timepulse |                                  | 0.25 Hz to 10 MHz           |
| Operational limits     | Dynamics<br>Altitude<br>Velocity | < 4g<br>80,000 m<br>500 m/s |
| Velocity accuracy      |                                  | 0.03 m/s                    |

Table 7. ZED-F20P performance in different GNSS mode

| GNSS                     | Parameter                                                   | GPS+GAL+BDS                                  | GPS+BDS | GPS+GAL | GPS |
|--------------------------|-------------------------------------------------------------|----------------------------------------------|---------|---------|-----|
| Acquisition              | Cold start<br>Hot start<br>Aided start                      | 25 s<br>2 s<br>2 s                           | -       | -       | -   |
| Update rate              | RTK<br>PVT<br>RAW                                           | 25 Hz<br>25 Hz<br>25 Hz                      | -       | -       | -   |
| Convergence time         | RTK                                                         | < 7 s                                        | -       | -       | -   |
| Horizontal pos. accuracy | PVT<br>RTK                                                  | 1.2 m CEP<br>0.006 m +<br>1ppm CEP           | -       | -       | -   |
| Vertical pos. accuracy   | RTK                                                         | 0.01 m +<br>1ppm CEP                         | -       | -       | -   |
| Sensitivity              | Tracking & Nav.<br>Reacquisition<br>Cold start<br>Hot start | -167 dBm<br>-160 dBm<br>-148 dBm<br>-158 dBm | -       | -       | -   |

Table 8. ZED-F20P supported GNSS constellations and signals

| System  | Signal                    | Frequency                                    |
|---------|---------------------------|----------------------------------------------|
| GPS     | L1C/A<br>L2C<br>L5        | 1575.420 MHz<br>1227.600 MHz<br>1176.450 MHz |
| Galileo | E1-B/C<br>E5a             | 1575.420 MHz<br>1176.450 MHz                 |
| BeiDou  | B1I<br>B1C<br>B2a         | 1561.098 MHz<br>1575.420 MHz<br>1176.450 MHz |
| QZSS    | L1C/A, L1C/B<br>L2C<br>L5 | 1575.420 MHz<br>1227.600 MHz<br>1176.450 MHz |
| NavIC   | L1-SPS Data<br>L5-SPS     | 1575.420 MHz<br>1176.450 MHz                 |

---

## 4. Applications

### 4.1 Standalone configuration

## 5. Communication interfaces

There are several communication interfaces, including UART, SPI, I2C, and USB. All inputs have internal pull-up resistors in normal operation and can be left open if not in use. All the PIOs are supplied by VCC, therefore all the voltage levels of the PIO pins are related to Vdd supply voltage.

### 5.1 UART Interface

There are two UART interfaces: UART1 and UART2. UART1 and UART2 operate up to and including a speed of 8 Mbaud. No hardware flow control on UART1 and UART2 is supported. UART1 is enabled by default if D\_SEL is not tied to ground.

Figure 3. UART chronogram specifications

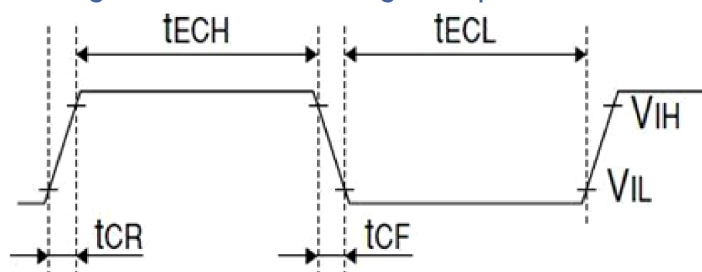


Table 9. ZED-F20P serial UART timing specifications

| Symbol    | Parameter                          | Min.                | Max.                | Units   |
|-----------|------------------------------------|---------------------|---------------------|---------|
| $V_{IL}$  | LOW-LEVEL input voltage            | 0                   | $0.2 \times V_{DD}$ | V       |
| $V_{IH}$  | HIGH-LEVEL input voltage           | $0.7 \times V_{DD}$ | $V_{DD} + 0.3$      | V       |
| $t_{ECH}$ | HIGH period of external data input | 0                   | 0.4                 | $\mu s$ |
| $t_{ECL}$ | LOW period of external data input  | TBA                 | TBA                 | $\mu s$ |
| $R_u$     | Baudrate                           | 4800                | 8                   | Mbps    |
| $t_{CR}$  | Rise time of data                  |                     | 5                   | ns      |
| $t_{CF}$  | Fall time of data                  |                     | 5                   | ns      |

## 5.2 SPI interface

The SP0701 has an SPI slave interface that can be selected by tying the **D\_SEL** input pin to ground. The SPI slave interface is shared with UART1. The SPI pins available are: SPI\_MISO (TXD), SPI\_MOSI (RXD), SPI\_CS\_N, SPI\_CLK. The SPI interface is designed to allow communication with a host CPU. The interface can be operated in slave mode only. Note that SPI is not available in the default configuration because its pins are shared with the UART and I2C interfaces. The maximum transfer rate using SPI is 950 kB/s and the maximum SPI clock frequency is 12.8 MHz.

This section provides SPI timing values for the ZED-F20P slave operation. The following tables present timing values under different capacitive loading conditions. The default SPI configuration is CPOL = 0 and CPHA = 0.

Figure 4. SPI chronogram specifications

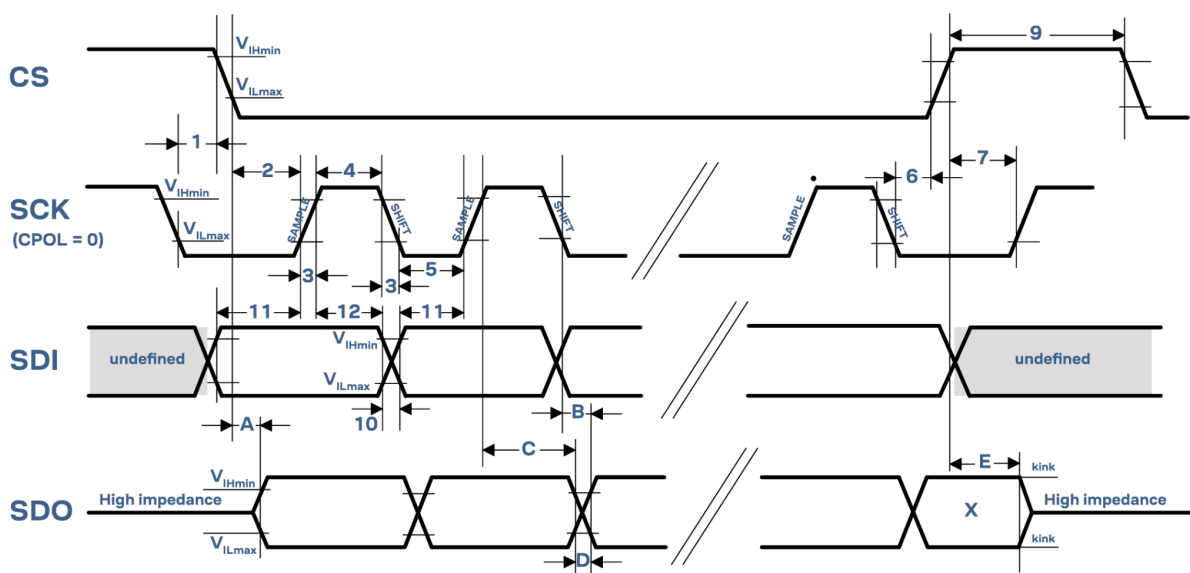


Table 10. ZED-F20P SPI timing specifications

| Timing value @ 20pF load                          | Min [ns] | Max [ns] |
|---------------------------------------------------|----------|----------|
| "A" - MISO data valid time (CS)                   | 17       | 42       |
| "B" - MISO data valid time (SCK) weak driver mode | 16       | 39       |
| "C" - MISO data hold time                         | 52       | 88       |
| "D" - MISO rise/fall time, weak driver mode       | 2        | 21       |
| "E" - MISO data disable lag time                  | 7        | 20       |

## 5.3 Slave I2C interface

An I2C compliant interface is available for communication with an external host CPU. The interface can be operated in slave mode only. It is fully compatible with Fast-Mode-Plus of the I2C industry standard. Since the maximum SCL clock frequency is 1 MHz (in Fast-Mode-Plus), the maximum bit rate is 1 Mbit/s. The interface stretches the clock when slowed down while serving interrupts, therefore the real bit rates may be slightly lower.

**Note :** The I2C interface is only available with the UART default mode. If the SPI interface is selected by tying the **D\_SEL** input to ground, the I2C is not available.

Figure 5. I2C chronogram specifications

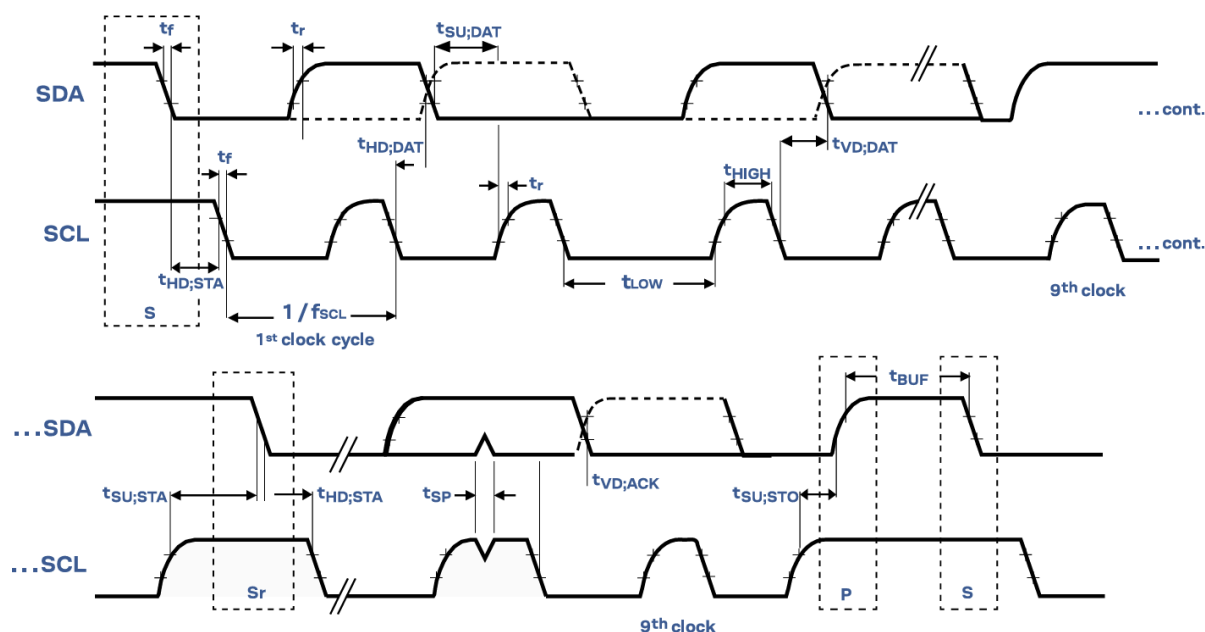


Table 11. SP0701 I2C timing specifications

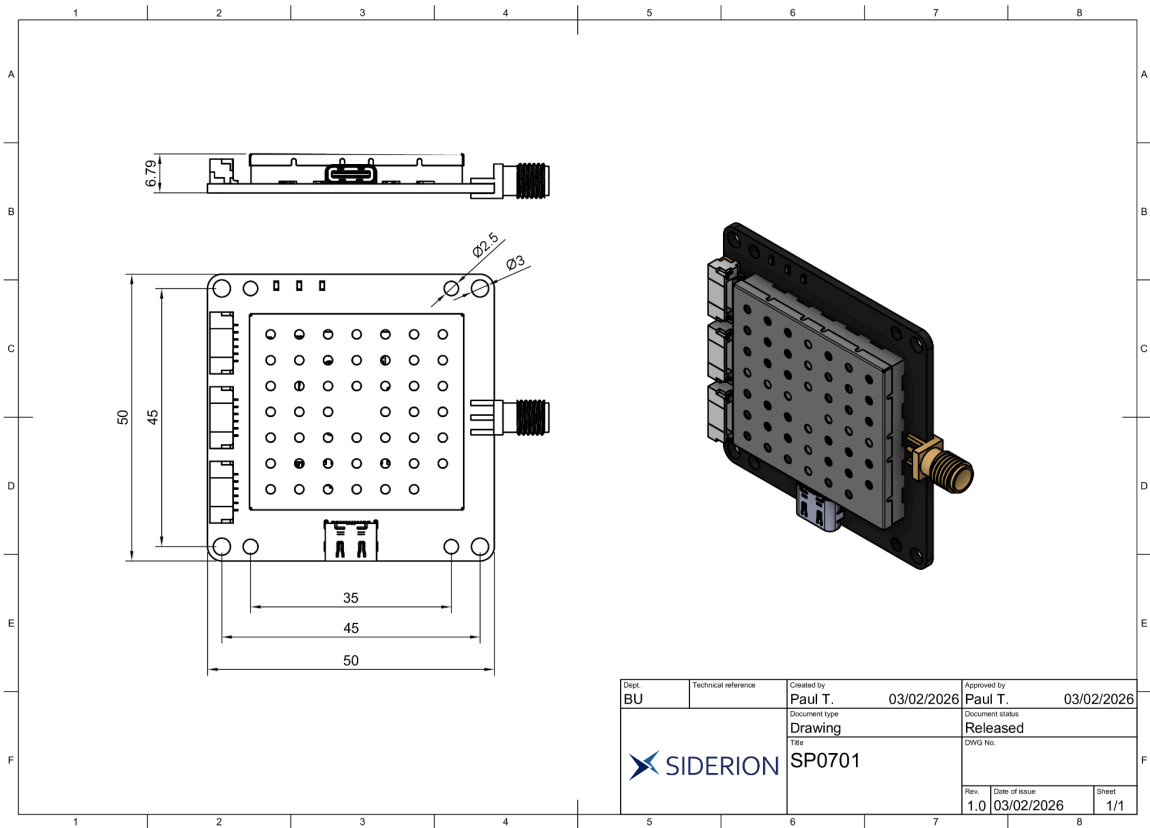
| Symbol | Parameter                         | Min                               | Max                        | Units |
|--------|-----------------------------------|-----------------------------------|----------------------------|-------|
| Vil    | LOW-LEVEL input voltage           | Vss-0.3                           | 0.3xVdd                    | V     |
| Vih    | HIGH-LEVEL input voltage          | 0.7xVdd                           | Vdd+0.3                    | V     |
| Cb     | Capacitive load for each bus line | 400<br>(standard &<br>fast modes) | 550<br>(fast mode<br>plus) | pF    |
| Fscl   | SCL clock frequency               | 0                                 | 1                          | MHz   |

## 5.4 USB interface

A USB interface, which is compatible with USB version 2.0 FS (Full Speed, 12 Mbit/s), can be used for communication as an alternative to the UART.

# 6. Mechanical drawings

Figure 6. SP0701 v1.0 mechanical drawings



## 7. Revision history

Table 12. Document revision history

| Date        | Revision | Changes                              |
|-------------|----------|--------------------------------------|
| 01-Sep-2025 | 1.0      | SiderionDoc_SP0701 / Initial release |
| 26-May-2026 | 1.1      | Figure 1 - Block Diagram added       |

## 8. Appendix

U-blox ZED-F20P datasheet: [ZED-F20P-00B Data sheet](#)

U-blox ZED-F20P integration manual: [ZED-F20P Integration manual](#)

U-blox ZED-F20P interface description: [u-blox F20 HPG 2.02 Interface description](#)

Siderion user's guide:

Siderion tutorials:

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